

Scale Invariant Feature Transform (SIFT)



Look in Relative Sense

"If you find a origin"

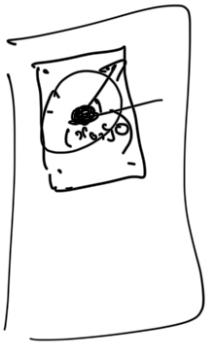
⇒ find an origin in a given image
⇒ Represent everything relative to the origin



SIFT Algorithm can be described in three steps: (All three steps have to be implemented for all blobs)

1) Find 'characteristic orientation of the blob. Blob center will become new origin (for the blob neighborhood)

2) Represent gradients of the neighborhood relative to new origin.
⇒ ("With respect to blob center ~~everything~~ All Coordinates are invariant)".



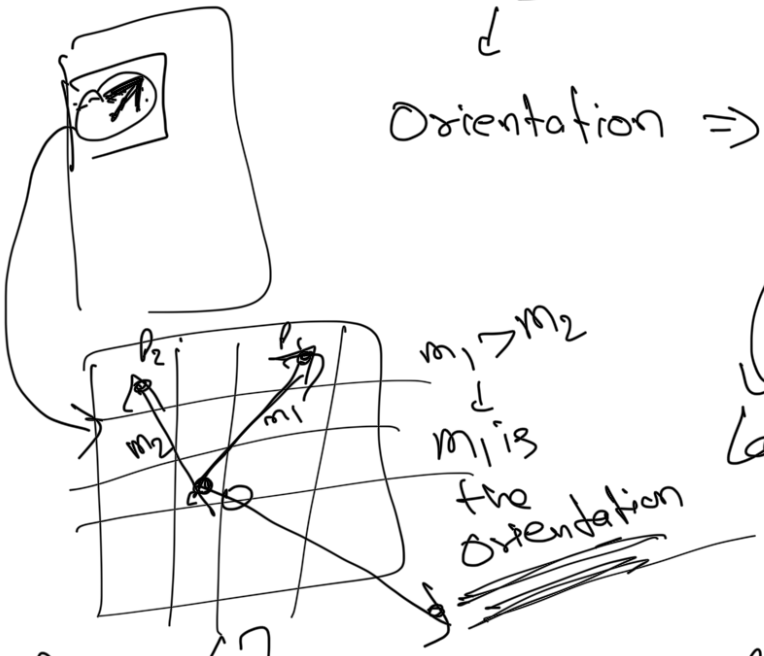
3) Normalize the gradients obtained in Step 2

The obtained vector is known as SIFT Descriptor.

Step1: Find orientation of the blob.

Orientation ⇒ Each pixel in the blob has a gradient wrt to blob center.

Largest gradient in the neighbourhood



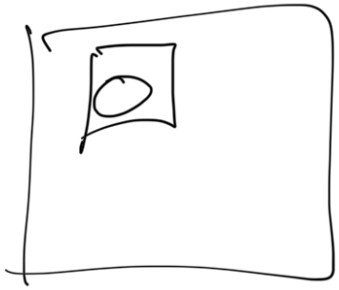
Use Gaussian weights for gradients Take

⇒ (x_0, y_0, σ_0)

Gaussian smoothed image at scale σ_0

$$I(x, y, \sigma) = L(x, y, \sigma) * I(x, y)$$

Take a neighbourhood ~~in the~~ of the blob



In the neighborhood

$$\frac{\partial I}{\partial x} = \frac{I(x+1, y) - I(x-1, y)}{2}$$

$$\frac{\partial I}{\partial y} = I(x, y+1) - I(x, y-1)$$

$\frac{\partial y}{\partial}$ 2

Sobel filter ← $I_x = \frac{\partial I}{\partial x} = I(x+1, y) - I(x-1, y)$

$I_y = \frac{\partial y I}{\partial y} = I(x, y+1) - I(x, y-1)$

magnitude ← $m(x, y) = \sqrt{I_x^2 + I_y^2}$

orientation ← $\theta(x, y) = \tan^{-1} \frac{|I_y|}{|I_x|} \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2})$

$\Rightarrow \tan^{-1} \frac{I_y}{I_x} \rightarrow (-\pi, \pi)$

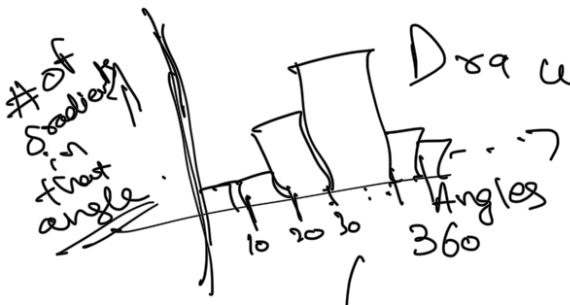
np.atan ← $\tan^{-1}$

np.atan2 takes sign of I_x and I_y into account

$\Rightarrow$ np.atan2



$\Rightarrow$ Get $m(x, y)$ and $\theta(x, y)$ for all points.



Draw a histogram.

David Lowe

Bin Size : 10^0

of Bins : 36

However ← Orientation is θ

where the histogram peaks.

30°

Keypoint (x_0, y_0, σ_0) → $(x_0, y_0, \sigma_0, \theta_0)$

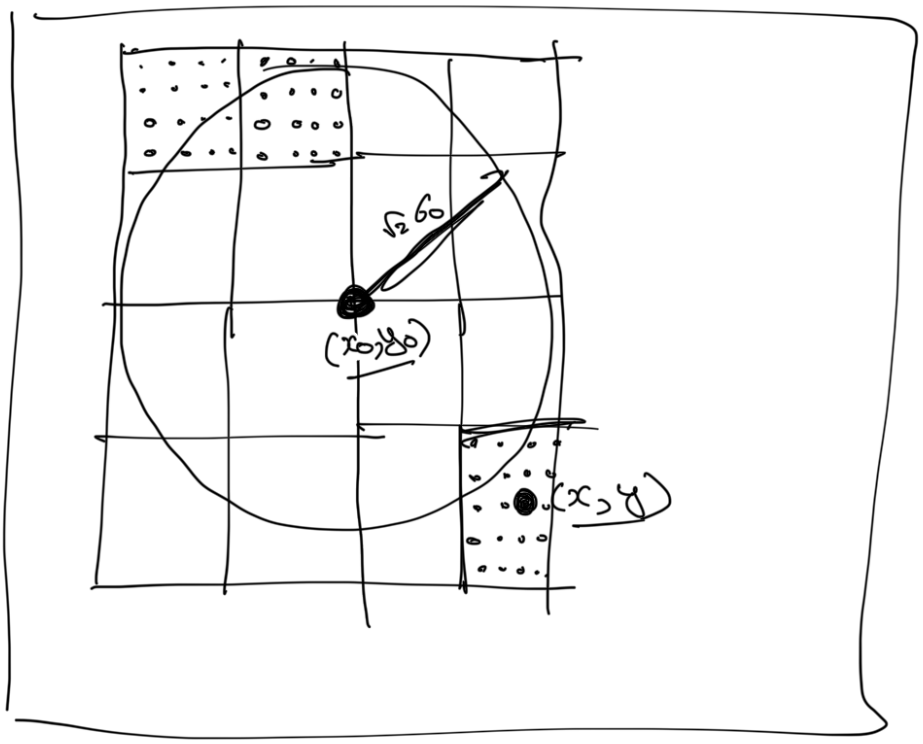
θ_0 → orientation of the blob

Step2:

Neighborhood

1) Encompasses the blob.

2) Divide the neighborhood ~~into~~ into 16 cells (where each cell is of size 4×4).



To find representation (or desired vector)

Represent each point in the neighborhood in a coordinate





system where (x_0, y_0) is the center and θ_0 orientation is the x-axis.

↓

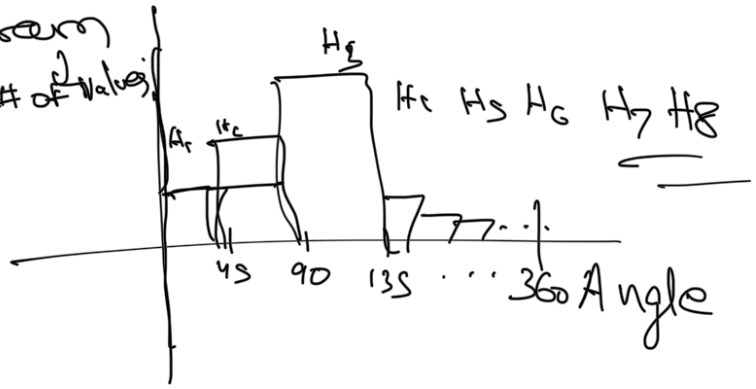
Every point in the neighbourhood is translated by (x_0, y_0) and rotated anti-clockwise by $-\theta_0$

Move every point (x, y) to (x', y') in new coordinate system where (x_0, y_0) is origin θ_0 is x-axis.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta_0 & \sin\theta_0 \\ -\sin\theta_0 & \cos\theta_0 \end{bmatrix} \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix}$$

⇒ For each cell find a 8 bin histogram

$\frac{360}{8} = 45°$



Each cell has ⇒ 8 values

16 cells will have ⇒ $16 \times 8 = 128$

Gradients of cells relative to the blob center. $\subseteq$ SIFT Descriptor

$$f = \begin{bmatrix} B_{11} \\ B_{21} \\ \vdots \\ B_{81} \\ B_{12} \\ B_{22} \\ \vdots \\ B_{82} \\ \vdots \\ B_{88} \end{bmatrix} \Rightarrow B_{(bin\#)(cell\#)}$$

$\Rightarrow$ Gradient may increase or decrease if illumination increase or decrease

$\Rightarrow$ Step3: Normalize the SIFT Descriptor

↙ illumination invariant.

$$\hat{f} = \frac{f}{\|f\|_2}$$

$\|f\|_2 = \sqrt{f_1^2 + f_2^2 + \dots f_{128}^2} \Rightarrow$

A large gradient should have a large effect.

↓ Gradient clipping

$$\hat{f}_i = \min(\hat{f}_i, 0.2)$$

Again Normalization is done to get unit Norm

SIFT descriptor

$$\hat{f}'' = \frac{\hat{f}'}{\|\hat{f}'\|_2}$$